

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Sixth Semester of B. Tech. (CL) Examination

May 2014

CL 310 Irrigation & Hydraulic Structures

Date: 15.05.2014, Thursday

Time: 10:00 a.m. To 01:00 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures with pencil only wherever required.
4. Use of scientific calculator is allowed.

SECTION – I

Q - 1 (a) Explain clearly the advantages and disadvantages of irrigation. [03]

(b) What are different surface methods of irrigation? Explain any ONE of them. [04]

Q - 2(a) Define the following terms: [05]

Culturable command area, Duty, Delta, Crop ratio, Field capacity

(b) Determine the design discharge of the canal for the following data: [06]

Crop	Base Period (days)	Area of crop (ha)	Duty at the head (ha/cumecs)
Sugarcane	365	250	650
Paddy (Kharif)	115	1800	750
Bajra (kharif)	110	2000	1200
Wheat (Rabi)	110	2500	1000
Vegetable(HW)	985	1500	650

Take time factor = 0.7, capacity factor = 0.8 and conveyance efficiency = 65 %

(c) What are different canal regulation works? What are the functions of distributory head regulator? [03]

OR

Q - 2 (a) Explain: Frequency of irrigation. What are the factors affecting it? Discuss any one of them in detail. [04]

(b) Find the field capacity of a farm soil for the following data: [05]

- (i) Root zone depth = 1.5 m
- (ii) Existing water content = 7 %
- (iii) Dry density of soil = 1350 kg/m³
- (iv) Water applied to soil = 600 m³
- (v) Water lost due to evaporation and deep percolation = 11 %
- (vi) Area of the land irrigated = 1100 m²

(c) Distinguish clearly between 'application efficiency' and 'storage efficiency'. [03]

(d) The surplus water escape is called the safety valve in canal system. Why? [02]

Q - 3 (a) Explain the following terms: [04]

(i) Weir (ii) SSHGL (iii) Head regulator (iv) Scouring sluices

(b) Explain Bligh's creep theory for the design of weirs on permeable foundation. [04]

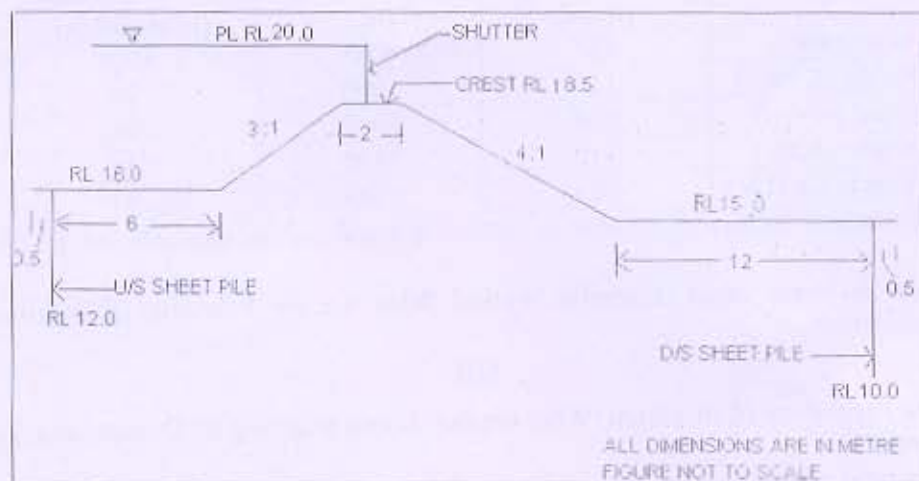
(c) What are the causes of failure of weirs founded on permeable soils? Also, suggest their remedial measures. [04]

(d) Explain, with suitable sketch, 'Superpassage'. [02]

OR

Q - 3(a) State different components of a diversion headwork and explain the functions of divide wall. [04]

(b) Obtain the residual uplift pressure heads and the thickness of the floor required at the key points for the weir shown in the figure, using Khosla's theory. The thicknesses of the floor at u/s end pile, intermediate pile and d/s end pile are assumed as 1m, 2m and 1.3m respectively. Take relative density of concrete as 2.4 and slope correction factor for 4:1 slope as 3.3. [08]



(c) Differentiate between Aqueduct and Siphon aqueduct. [02]

SECTION – II

- Q - 4 Design an unlined irrigation channel for non-alluvial soil using Chezy's formula for [07]
data given below:

$Q = 45$ cumecs, Mean velocity = 0.9 m/sec, $B = 5D$, Side slope = $1:1$, $K = 1.2$

- Q - 5 Attempt any Two:

[14]

- (a) Describe with neat sketches the various methods adopted for controlling seepage through the body and foundation of the earthen dam.

- (b) Design and draw a low gravity dam for the following data:

Height of low dam = 88.2 m

Specific gravity of concrete = 2.4

Unit weight of water = 10 kN/m³

- (c) What are the various modes of failures of a gravity dam? Explain any TWO of them.

- Q - 6 (a) Distinguish between elementary profile and practical profile with neat sketch of each. [02]

- (b) Attempt any Two:

[12]

- 1 Define the following with neat sketch:

(i) Corbel in spillway (ii) Different relative positions of TWRC and JHC

- 2 What is meant by an energy dissipator? Discuss the various methods used for energy dissipation below spillways.

- 3 Define: Spillway. What is the purpose to provide it? What are the essential requirements? Where the spillway is located?
